ONBRD-02: IAM and Authentication

> **Series:** ONBRD — Dynatrace Onboarding | **Notebook:** 2 of 10 | **Created:** December 2025 | **Last Updated:** 11/11/2026

Setting Up Secure Access

Before inviting your team, configure authentication and permissions properly. This notebook covers SAML/SSO setup, API tokens, and the modern permission model.

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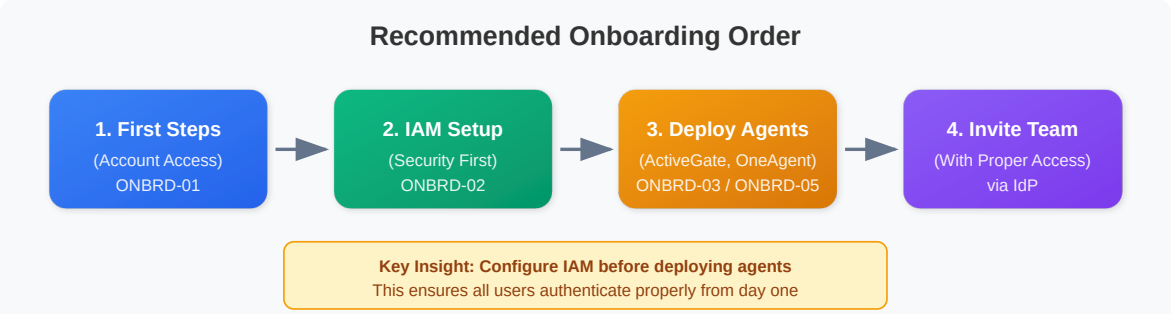
Prerequisites

- Account owner or admin access
- Identity provider details (for SSO)
- Understanding of your organization's access requirements

1. Why IAM First?

Setting up IAM before deploying OneAgent or inviting users ensures:

Benefit	Why It Matters
Consistent access	Users authenticate the same way from day one
Proper permissions	No accidental admin access for viewers
Audit trail	All access tied to corporate identity
Offboarding	With SAML, a disabled IdP account can't sign in; with SAML + SCIM, users are also removed when deprovisioned in the IdP
Compliance	Meet security requirements from the start



> **Sources:** Enterprise identity management (DT docs) — *"Users are removed immediately when deprovisioned in the IdP"* (SAML + SCIM).

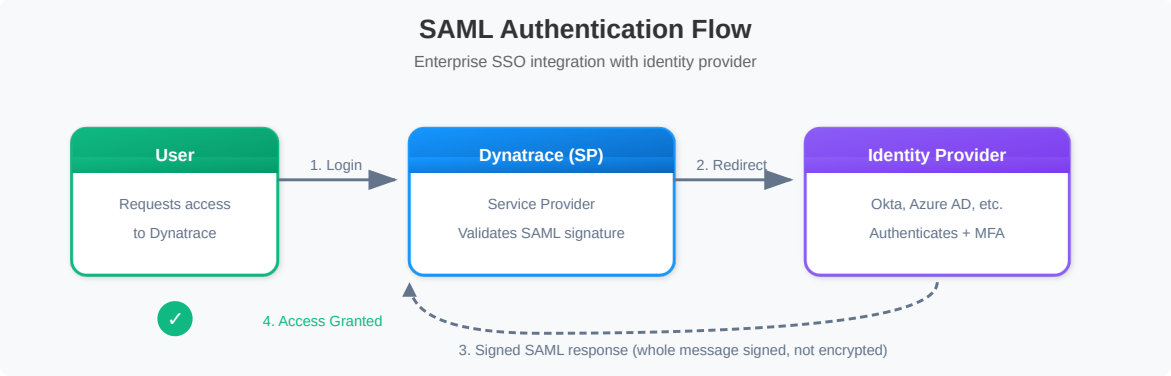
2. Authentication Options

Dynatrace manages users either in its own user database or through an IdP using the federation protocols **SAML** and **SCIM**:

Method	Description	Best For
Local Users	Built-in user accounts	Small teams, testing, the non-federated fallback account

Method	Description	Best For
SAML 2.0	Delegated authentication (SSO) through your IdP	Most organizations
SCIM (with SAML)	Automated provisioning and deprovisioning of users and groups from your IdP	Dynatrace's recommended enterprise approach for most organizations

Authentication Flow



Common Identity Providers

- **Azure Active Directory (Entra ID)** - Microsoft environments
- **Okta** - Cloud-native identity
- **OneLogin** - Enterprise identity
- **PingFederate** - On-premises/hybrid
- **Google Workspace** - Google-centric organizations

> **Sources:** Identity management (DT docs) — *"an IdP with the supported federation protocols SAML and SCIM"*; Enterprise identity management (DT docs) — *"This is the recommended enterprise approach for most organizations."*; SAML (DT docs) — *"No assertion encryption."*

3. Configuring SAML SSO

Dynatrace documents four steps, in this order: create a fallback user account, verify your domain, configure SAML, test.

Step 1: Create a Fallback User Account

Location: Account Management → Identity & access management → User management → **Invite user**

Invite a **non-federated** user — an email address on a *different* domain from the one you are federating — and add it to a group with the **View and manage users and groups** permission. This account is how you get back in if the SAML configuration locks you out.

Step 2: Verify Domain Ownership

Location: Account Management → Identity & access management → **Domain verification**

1. Add the domain (for example `mycompanyname.com`); add every domain your users sign in with
2. Copy the TXT resource record and add it to the domain's DNS
3. Select **Actions** → **Verify** (DNS propagation can take minutes, occasionally up to 24 hours)

Step 3: Configure SAML

Location: Account Management → Identity & access management → **SAML configuration** → **New configuration**

1. Choose the **federation type** (Global, Account, or Environment)
2. Download the Dynatrace **SP metadata** (Entity ID, Assertion Consumer Service URL, Logout URL) and register it in your IdP
3. In your IdP, configure the Dynatrace application so that:
   - The **entire SAML message is signed** — signing only the assertion is rejected with `400 Bad Request`
   - The assertion is **not encrypted**
   - The NameID format is `urn:oasis:names:tc:SAML:1.1:nameid-format:emailAddress` (the user's email comes from the NameID)
4. Upload the IdP metadata XML to Dynatrace
5. *(Optional)* Map attributes: first name, last name, and the federated group/role attribute
6. Select the domains, activate the configuration (**Enable SSO**), and complete it

Step 4: Test SSO

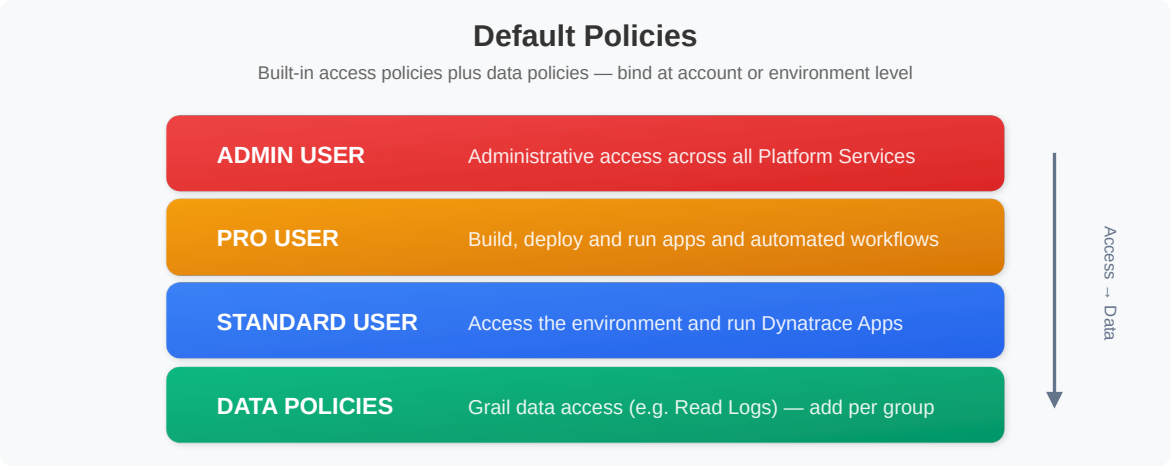
1. Open a new browser instance and a new incognito/private window
2. For Account or Environment federation, go to your tenant URL first (`https://{tenant-id}.apps.dynatrace.com`) — that is how Dynatrace SSO gets the context; sign-in is routed by your email domain
3. Complete IdP authentication
4. Verify you land in Dynatrace with the correct permissions, and that you can still see **User management** and **Group management**

> **Warning:** If sign-in breaks, use the fallback account from Step 1 to fix or disable the configuration. SAML sessions time out after 1 hour; that timeout is not configurable.
>
> **Sources:** SAML (DT docs) — *"The entire SAML message must be signed (signing only SAML assertions is insufficient and generates a 400 Bad Request response)."*; *"Your fallback account must be a non-federated user account belonging to a group that has View and manage users and groups permission and isn't covered by the federated sign-in."*; *"To test an Account or Environment federation, go to the tenant URL first"*.

4. User Groups and Permissions

Permission Model

Dynatrace uses policy-based access control. Its default policies come in two types — **access policies** (built-in user roles) and **data policies** (access to Grail data) — and you assign them to groups at the account level (all environments) or for an individual environment:



Recommended Groups

A suggested starting mapping (adapt it to your organization):

Group	Default policies	Use Case
Platform Admins	Admin User	Platform team, IAM management
SRE Team	Pro User + data policies	Workflow setup, monitoring config
Developers	Standard User + scoped data policies	Problem response, dashboards
Stakeholders	Standard User + read-only data policy	Reports, read-only access

Creating Groups

Location: Account Management → Identity & access management → Group management

1. Click "Create group"
2. Name the group (e.g., "SRE-Team")
3. Add a description
4. Assign policies (permissions)
5. Optionally link to IdP groups for automatic membership

Scoping Access with Policies

The modern platform uses **policies** to control what users can access:

Policy Type	Purpose
Access policies	Feature-level access (Admin User, Pro User, Standard User, or your own)
Data policies	Access to Grail data, scoped by bucket / table / `storage:dt.security_context` conditions. (Segments filter views; they don't grant access.)

Either type is bound at the **account** level (all environments) or the **environment** level.

Parameterized Policies (Strongly Recommended)

Prefer **one parameterized policy bound to multiple groups via binding parameters** over many copies of the same policy with hardcoded scope values:

```
ALLOW storage:buckets:read WHERE storage:table-name = "logs";
ALLOW storage:logs:read WHERE storage:dt.security_context =
"${bindParam:team}";
```

The first statement is not optional: a table permission alone returns no records. `storage:buckets:read` is *"Required additionally to a table permission."*

Parameters are written as `${bindParam:<name>}` — Dynatrace's policy templating rule is *"Policy parameters should be prefixed with bindParam: and enclosed in ${...}."* (Policy templating (DT docs)). Bind the policy to `team-payments`, `team-checkout`, `team-fraud`, etc., varying only the `team` parameter — one policy, N bindings. The parameter shape is load-bearing: design once, change rarely.

The **`dt.security_context`** field is the standardized boundary for Gen3 IAM scoping across both data and configurations. Without it, cross-entity-type policies cannot be written. Decide your `dt.security_context` value space before tagging anything (covered in **ONBRD-06**).

Where to Go Deeper

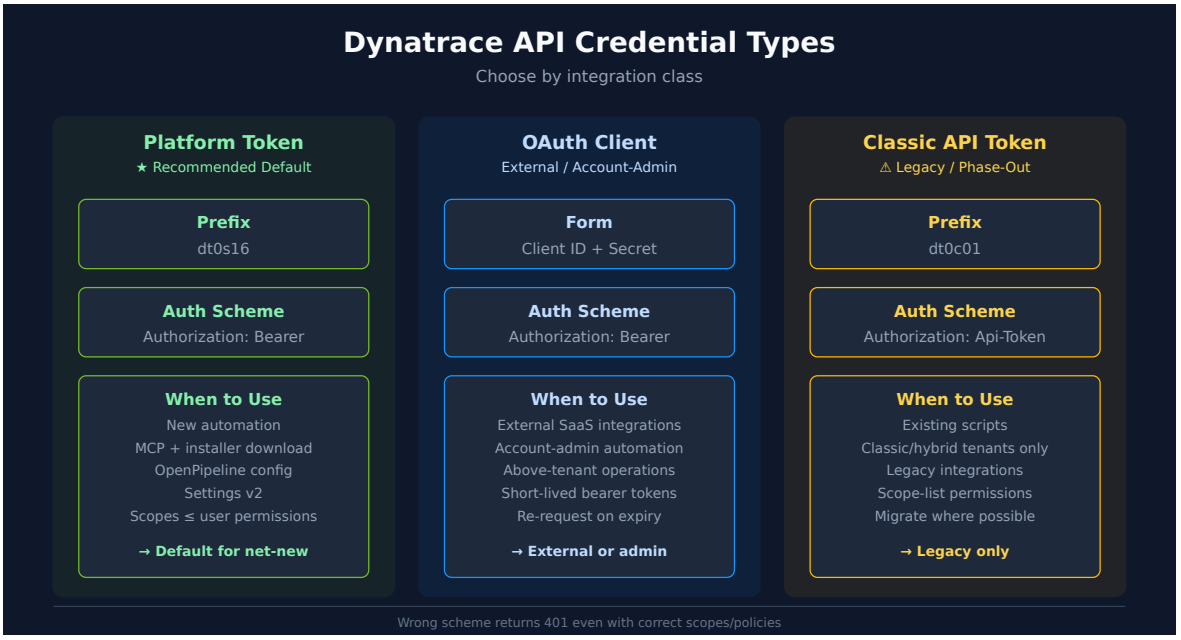
- **IAM-04 / IAM-05** — Designing effective policies and boundary conditions

- **IAM-10: Templated Policy-Group Assignments** — binding one templated policy to many groups with `bindParam` values
- **IAM-11 (WORKSHOP)** — Hands-on policy and persona design
- **IAM-99** — IAM best-practice summary and DQL reference
- **FAQ-02** — Tagging sources, standards, and strategy (`dt.security_context` design)
- **ORGNZ-06 / ORGNZ-07** — Security context for data and configuration scoping

> **Note:** For data filtering, use **Segments +** `dt.security_context` (covered in ONBRD-06) rather than the legacy Management Zones. Segments narrow what a user sees; access itself is enforced by IAM policies.
>
> **Sources:**
>
> - Default policies (DT docs) — *"You can assign policies to groups via the user group details either on the account level, which includes all environments in that account, or on the individual environment level."*
> - IAM policy statements (DT docs) — *"Grants permission to read records from Grail buckets. Required additionally to a table permission."*
> - Segments visibility (DT docs) — *"All queries, with or without segments, always respect data access permissions enforced by IAM policies."*

5. API Token and OAuth Management

The modern platform supports three credential types — pick based on the integration:



Credential	Prefix / Form	When to Use
Platform Token *(recommended default for new automation)*	`dt0s16`	New automation, MCP integrations, OpenPipeline configuration, Settings v2, installer downloads
OAuth 2.0 Client	client ID + secret	External SaaS integrations, account-admin automation
Classic API Token *(legacy phase-out)*	`dt0c01`	Existing scripts on classic/hybrid environments; migrate to Platform Token where possible

> **Workflows don't use a token.** Every workflow task runs as the workflow's **actor** (a user or service user; by default the workflow's creator) — grant permissions to the actor, not a token.

Platform Tokens (Recommended Default)

Platform Tokens are the standard for new automation — in Latest Dynatrace they replace classic access tokens. They:

- Carry the **scopes you select** at creation, and only work within the permissions of the user (or service user) they are issued for — *"A platform token will only work within the limits of the assigned user's permissions."* (Platform tokens (DT docs))

- Are bound to a single user identity (or service identity) for traceability
- Should be the default unless you have a specific reason to choose OAuth or Classic

Location: My platform tokens (`myaccount.dynatrace.com/platformTokens`) for your own tokens; admins: **Account Management** → **Identity & access management** → **Platform tokens**

```
# Authorization scheme by token prefix:
#   dt0s16          →  Authorization: Bearer &lt;token&gt;  (dt0s01 is a
separate SCIM account token, not a Platform Token)
#   dt0c01          →  Authorization: Api-Token &lt;token&gt;
# Wrong scheme returns 401 even with correct scopes/policies.
```

OAuth 2.0 Clients

Use for **external system integrations** or **account-admin automation** that operates above the tenant level.

Location: Account Management → Identity & access management → OAuth clients

1. Create an OAuth client
2. Use client credentials flow to obtain bearer tokens
3. Bearer tokens are short-lived (the documented example response has `"expires_in": 300` , in seconds) — request a new one with the client credentials when it expires

Classic API Tokens (Legacy)

Classic access tokens exist only on classic and hybrid environments — *"Classic access tokens don't exist in latest environments"* — so existing scripts that use them need a migration plan. Migrate to Platform Tokens during routine refresh cycles. Classic tokens use the `Api-Token` Authorization scheme (not `Bearer`).

Location: the **Access Tokens** app in your environment → **Generate new token** (not Account Management)

Common Use	Classic scope	Platform-token scope
OneAgent Installer Download	`InstallerDownload`	`fleet-management:oneagents:download`
ActiveGate Installer Download	`InstallerDownload`	`fleet-management:activegates:download`

Common Use	Classic scope	Platform-token scope
Metric Ingestion	`metrics.ingest`	—
Log Ingestion	`logs.ingest`	—

Token Best Practices

Practice	Why
Minimal scope / least-privilege policy	Limit blast radius if compromised
Descriptive names	Know what each token is for
Expiration dates	Force rotation, reduce risk
Separate tokens per use	Revoke one without affecting others
Never commit to code	Use environment variables or secrets managers
Prefer Platform Tokens for new work	Aligned with the Gen3 IAM model

Where to Go Deeper

- **IAM series** — Token management, lifecycle, audit patterns

> **Sources:**
>
> - Upgrade from access tokens classic (DT docs) — *"In Latest Dynatrace , this model is replaced with platform tokens"*
> - Download latest OneAgent installer (DT docs) — *"Platform Token / OAuth: Required scope: fleet-management:oneagents:download"*
> - Download latest ActiveGate installer (DT docs) — *"Platform Token / OAuth: Required scope: fleet-management:activegates:download"*
> - Workflow security (DT docs) — *"By default, the actor is the creator of the workflow."*
> - OAuth clients (DT docs), API authentication — token prefixes (DT docs)

6. Verification Queries

After configuring IAM, these queries confirm that people are signing in to this environment. They read environment audit events in `dt.system.events`.

> **One sign-in can be two events.** From SaaS 1.349 (rollout start 10/27/2026), sign-in and sign-out audit events reach Grail, and *"signing in to a classic app can produce two audit events—one from the API gateway and one from the cluster"*. Dynatrace says *"You can differentiate them by their event provider"*, so both queries below show `event.provider` (on the validation tenant, 11/11/2026: `API_GATEWAY` and `CLASSIC_LOGIN`, 91 and 12 events over 7 days). Count one provider when you need a number of sign-ins rather than a number of events. (SaaS 1.349 release notes (DT docs))
>
> **What they can't show:** SSO configuration, group and permission changes are recorded in the separate **Account Management audit log** (Account Management UI, or the Account Audits API with the `account-audit-logs-read` scope) — *"Neither v2/auditlogs nor dt.system.events captures it."* (Upgrade from audit logs classic (DT docs)). Use the manual checklist below to confirm SSO itself.

```
// Environment sign-ins (platform gateway audit events)
// A classic-app sign-in can log twice (API_GATEWAY and CLASSIC_LOGIN) - read
event.provider (SaaS 1.349).
// Data object corrected 09/24/2026. The Dynatrace audit trail is NOT in
`logs`: the former
// `fetch logs | filter matchesPhrase(log.source, "audit")` matched nothing,
or matched an
// unrelated file-based audit log (a database .aud file on the validation
tenant). Environment
// audit records are structured events in `dt.system.events` with event.kind
== "AUDIT_EVENT".
fetch dt.system.events, from:-24h
| filter event.kind == "AUDIT_EVENT"
| filter event.type == "LOGIN"
| fields timestamp, user.id, event.provider, event.outcome,
authentication.type, origin.type
| sort timestamp desc
| limit 50
```

```
// Sign-in outcomes over the last 7 days (audit events, not application logs)
// Split by event.provider: a classic-app sign-in can produce two events, one
per provider (SaaS 1.349).
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter event.type == "LOGIN"
| summarize {events = count(), users = countDistinct(user.id)}, by:
{event.provider, event.outcome}
| sort events desc
```

Manual Verification Checklist

Item	How to Verify
SSO working	Login via IdP in incognito window
Groups created	Check Account Management → Identity & access management → Group management
Permissions assigned	Test with a viewer account
Fallback account	The non-federated fallback user can still sign in
API tokens	Installer-download token ready (platform token with `fleet-management:oneagents:download`, or classic `InstallerDownload` on classic/hybrid tenants)

7. Next Steps

With IAM configured, you're ready to:

1. **ONBRD-03: Deploying ActiveGate** — Set up network routing (if needed for restricted networks)
2. **ONBRD-04: Cloud & SaaS Integrations** — Connect AWS / Azure / GCP and SaaS sources
3. **ONBRD-05: Deploying OneAgent** — Start collecting infrastructure and application data
4. **ONBRD-06: Organizing Your Environment** — Set up tags, segments, and `dt.security_context`

5. Invite team members via your IdP
6. Create additional Platform Tokens or OAuth clients as needed

IAM Tasks Before Moving On

- [ ] SAML/SSO configured and tested
- [ ] Non-federated fallback account created and documented
- [ ] User groups created for major roles
- [ ] Parameterized policy shape decided; group → policy bindings drafted
- [ ] `dt.security_context` value space planned (deeper in ONBRD-06 / FAQ-02)
- [ ] Platform Token issued for migration / automation tooling
- [ ] Installer-download token issued (platform token with `fleet-management:oneagents:download`, or classic `InstallerDownload` on classic/hybrid tenants)
- [ ] Token naming conventions established

Where to Go Deeper

- **IAM series** (15 notebooks) — full IAM administration depth
- **FAQ-02** — Tagging sources, standards, and strategy
- **ORGNZ series** — Bucket strategy, segments, security context

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Summary

In this notebook, you learned:

- Why IAM should be configured before deploying agents
- Authentication options (Local, SAML, SCIM)
- How to configure SAML SSO
- Default policies (Admin User, Pro User, Standard User, data policies), group structure, and parameterized policies
- The three credential types (Platform Token, OAuth, Classic API Token) and when to use each
- `dt.security_context` as the standardized boundary field for Gen3 IAM

- How to verify IAM configuration

References

- Identity and access management (DT docs)
- Platform tokens (DT docs)
- Policy templating (DT docs) — the `${bindParam:...}` syntax quoted in section 4
- SAML (DT docs) — fallback account, domain verification, IdP requirements
- SAML configurations (DT docs)
- Enterprise identity management — SAML and SCIM (DT docs)
- Default policies (DT docs)
- Identity management — SSO and federation (DT docs)
- Access tokens classic (DT docs)
- OAuth clients (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.